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MILITARY SPECIFICATION

POWER SUPPLY PP-6224( )/U

This specification is approved for use by the US Army Electronics Research & Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high efficiency power supply designated Power Supply PP-6224( )/U for furnishing 0-25 amperes over a voltage range of 24 to 32 Vdc from nominal 115/230 Vac, 50, 60, or 400 Hz, single phase power sources.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Packaging, Methods of
MIL-T-152	Treatment, Moisture-and Fungus-Resistant of Communications, Electronic and Associated Electrical Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications & Electronics Materiel Readiness Command, ATTN: DRSEL-LE-ST, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

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MIL-S-901	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements For (Navy)
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
DL-SM-B-667960	Power Supply PP-6224( )/U

SPECIAL PACKAGING INSTRUCTIONS

CERCOM

SPI 1G0001	Special Packaging Instructions

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with Drawing DL-SM-B-667960 and the requirements of this specification.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Printed wiring boards. Printed wiring boards used in the construction of printed wiring assemblies, shall be a product which has passed the requirements of MIL-P-55110, MIL-STD-275 and the master drawing. However, those parts of MIL-STD-275 which are in conflict with the applicable printed wiring board drawings and which require redesign of the printed wiring boards shall not apply. Conflicts between master drawings and MIL-STD-275 need not be submitted as required by MIL-P-55110.

3.4.1 Protective coating. Printed wiring assemblies shall be protected on both sides of the printed wiring assembly with conformal coating in accordance with MIL-I-46058 as required by MIL-STD-275.

3.5 Marking. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial number. Each Power Supply PP-6224( )/U shall be serial numbered.

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3.6 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.6)

3.7 Electrical requirements.

3.7.1 Input voltage. The power supply shall deliver the output required by this specification (See 3.7.2) from the following single phase input sources:

Volts ac (±10%)	Frequency - Hz (±5%)
115/230	50, 60 and 400

3.7.2 Output voltage. The voltage across the output of the power supply shall be 28.0 volts (±1%) over the load range of 0-25 amperes. The output voltage shall also be adjustable over the range of 24.0 to 32.0 volts by an external locking type control and shall be within ±1% of the voltage setting over the load range from 0 to 25 amperes. (See 4.7.1.)

3.7.3 Ripple. The rms ripple voltage shall not exceed 0.5 per cent at any output condition from no-load to full-load. The peak to peak ripple voltage including switching spikes and other noise components shall not exceed ±1% at any output condition from no-load to full-load. The ripple voltage includes voltage perturbations within the frequency range of 20 Hz to 10 MHz. (See 4.7.2)

3.7.4 Recovery time. The output voltage of the power supply shall recover to 28.0 volts ±1% within 25 milliseconds when the load is switched from no-load to full-load and from full-load to no-load. The maximum instantaneous overshoot voltage under these conditions and for any turn on or turn off condition of the power supply shall not exceed 20% of the nominal voltage (28.0 Vdc). (See 4.7.3)

3.7.5 Overload protection. The power supply shall be protected from physical and electrical damage from any overload conditions (including a direct short circuit) by the use of circuit breakers in the "ac input" circuit and the "dc output" circuit, and by automatic current limiting circuitry. (See 4.7.4)

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3.7.6 Power changeover. The power supply shall include an under voltage relay which shall automatically transfer the "OUTPUT" connector to receive power from the "DC INPUT" connector when the 115 Vac input is lost or varies to a low of 77 (plus 12, minus 0) Vac or the 230 volt ac input is lost or varies to a low of 154 (plus 24, minus 0) Vac. This condition shall be indicated by a red light on the front panel labelled "DC EMER" and extinguishing of the "DC NORM" green light. The dc power supply shall continue to deliver power at an output voltage of no less than 19 volts at the minimum input changeover voltage. The relay shall return the "OUTPUT" connector circuit to normal when the 115 volt ac input is restored or rises to 103 (plus 0, minus 12) Vac, or if the 230 volt ac input is restored or rises to 206 (plus 0, minus 24) Vac. The normal condition shall be indicated by extinguishing of the "DC EMER" red light and lighting of the "DC NORM" green light. There shall be consistency of polarity of the circuit connecting the "battery in" plug and the "power out" plug. (See 4.7.5)

3.7.7 Current limiting. Automatic control circuitry shall limit the output current to a maximum value of 30 amperes to protect the power supply from overload and automatically restore the output when the overload is removed. (See 4.7.4)

3.7.8 Protection.

3.7.8.1 Battery discharge. The power supply when used as a battery charger shall include provisions to prohibit the discharge of the battery being charged under the conditions of:

a. Loss of power input to the charger.
b. Battery emf in excess of dc output voltage.

3.7.8.2 General. The power flow conversion circuitry and feedback control circuitry of the power supply shall be fully protected under operational and abnormal conditions which include the following:

a. Short circuit of dc output.
b. Reverse polarity of the battery at the output.
c. Progressive and prolonged output overloads.
d. Over voltage line surges and transient voltage spikes.
e. Under voltage of the input voltage source below the minimum specified.

The equipment shall automatically resume normal operation when any of the above faults are removed. (See 4.7.6 and 4.7.11)

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3.7.9 Efficiency. The overall efficiency of the power supply shall not be less than 70%. (See 4.7.8)

3.7.10 Remote sensing. The power supply shall include provisions for extending the sensing of the dc output voltage through signal leads from the auxiliary output receptacle to the remote load terminals. Also, provisions shall be included to automatically revert to local sensing if the remote circuit is inadvertently opened. (See 4.7.7)

3.7.11 Stability. Output voltage drift at a constant load and ambient temperature shall not exceed 0.5% over a period of 24 hours. (See 4.7.9)

3.7.12 Temperature coefficient. Change in output voltage as a function of ambient temperature shall not exceed 0.02 percent per degree F (0.036 percent per degree C) over the entire specified operating temperature range. (See 4.7.10)

3.7.13 Bleeder network. The bleeder network shall discharge the AC line filter so that the voltage at the input connector pins drops to 30 volts or less within two seconds after the power is turned off.

3.8 Environmental conditions. The equipment shall meet the following environmental conditions:

3.8.1 Temperature. The equipment shall operate at ambient temperature in the range of +150°F (+66°C) to -40°F (-40°C). (The 150°F (66°C) temperature includes effect of sunload.) Exposure during operation at the high temperature extreme shall not exceed 4 hours and at the low temperature extreme shall not exceed 72 hours, at any one time. The equipment (non-operating) shall withstand exposure in the range of +160°F (+71°C) to -80°F (-62.2°C), exposure at the high temperature not to exceed 48 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.8.1.1 Low. Before, during, and after testing as specified in 4.8.1.1, the equipment shall meet full specification performance for the following measurements:

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Measurement	Requirement Paragraph	Test Paragraph
Input and Output Voltage	3.7.1, 3.7.2	4.7.1
Ripple Voltage	3.7.3	4.7.2
Recovery Time	3.7.4	4.7.3
Power Changeover	3.7.6	4.7.5

3.8.1.2 High. Before, during, and after testing as specified in 4.8.1.2, the equipment shall meet full specification performance for the following measurements:

Measurement	Requirement Paragraph	Test Paragraph
Input and Output Voltage	3.7.1, 3.7.2	4.7.1
Ripple Voltage	3.7.3	4.7.2
Recovery Time	3.7.4	4.7.3
Power Changeover	3.7.6	4.7.5

3.8.2 Relative humidity. The equipment shall withstand up to 98 percent relative humidity for ten days. When tested as specified in 4.8.2, the PP-6224( )/U shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet full specification performance during the specified operating times for the following measurements:

Measurement	Requirement Paragraph	Test Paragraph
Input and Output Voltage	3.7.1, 3.7.2	4.7.1
Ripple Voltage	3.7.3	4.7.2

3.8.3 Altitude (elevation). The equipment shall operate up to 15,000 feet above sea level and shall withstand storage up to 40,000 feet above sea level. Full specification performance for the following measurements shall be met before, during and after testing as specified in 4.8.3:

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Measurement	Requirement Paragraph	Test Paragraph
Input and Output Voltage	3.7.1, 3.7.2	4.7.1
Ripple Voltage	3.7.3	4.7.2

3.8.4 Leakage (immersion). The equipment shall show no evidence of leakage after it is tested as specified in 4.8.4.

3.8.5 Vibration, bounce and shock. The equipment shall meet the requirements of Table I.

TABLE I - Vibration, bounce and shock

Inspection	Insp. Para.	Performance after test
Vibration	4.8.5	Specified performance 1/
Bounce test	4.8.6	Specified performance 1/ No physical damage
Shock test; ballistic	4.8.7	Specified performance 1/ Any physical damage shall be minor only.

1/ Prior to and after testing, the equipment shall meet specified performance for the following requirements: 3.7.1, 3.7.2, 3.7.3 and 3.7.6.

3.8.6 Fungus. After the fungus test of 4.8.8, the PP-6224( )/U shall be visually examined, using a 10 power magnifier. The PP-6224( )/U shall show no more than .... minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (See 6.8.3) as evidenced by growth colonization (See 6.8.2) which includes branching (See 6.8.1) and sporulation (See 6.8.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (See 6.8.5) shall not to included in this evaluation.

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3.8.7 Blowing sand and dust. The equipment shall withstand the following:

a. Blowing sand. Particle size .01 to 1.0 mm with wind speeds of 15 knots or greater.

b. Blowing dust. Particle size .0001 to .01 mm with wind speeds of 15 knots or greater.

Before and after testing as specified in 4.8.9, the equipment shall meet full specification performance for the following measurements:

Measurement	Requirement Paragraph	Test Paragraph
Input and Output Voltage	3.7.1, 3.7.2	4.7.1
Ripple Voltage	3.7.3	4.7.2

3.9 Electromagnetic Interference (EMI). The equipment shall comply with the following emissions and susceptibility requirements of Notice 4 of MIL-STD-461. (See 4.11)

CE01	CS02	RE02	RS03
CE02		RE02.1	
CE04			

RE02.1 shall be performed from 14 KHz to 1 GHz. RS03 shall be performed from 10 KHz to 400 MHz as follows:

Frequency Range	Susceptibility Level
10 KHz to 1.99 MHz	1 Volt/meter
2.0 MHz to 29.99 MHz	5 Volt/meter
30 MHz to 400 MHz	10 Volt/meter

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3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9) Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.11 Preconditioning. The equipment shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1. (Also see 4.5)

3.12 Burn-in. Each equipment shall be subjected to a burn-in period with 24 hours of continuous failure-free on-time in accordance with 4.6.2.

3.13 Tropicalization. Parts and assembled equipment shall be treated in accordance with MIL-T-152. Prior to treatment, the supplier shall submit for approval to the Government, a detailed statement of treatment as required by MIL-T-152. (See 6.2.f)

3.14 Air-seal test, provision for. The case shall provide means for enabling performance of the air-seal test. (See 4.8.10 and 6.7)

3.15 Reliability. The power supply shall have a MTBF (0_o) of 5000 hours minimum when operated under the following conditions:

a. Ambient temperature of 122°F (50°C).
b. Duty cycle-intermittent operation.
c. Environment-ground mobile.

The above requirement shall be demonstrated in accordance with 4.12.

3.16 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

a. In MIL-P-11268 (including specified requirements of MIL-STD-454):

(1) Tropicalization.
(2) General requirements for plastic material and plastic parts.

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(3) Mechanical operation of controls.
(4) Printed wiring assemblies.
(5) Screws other threaded devices and related parts.
(6) Cleaning.
(7) Riveting.
(8) Process for solder (including flux and cleaning agents).
(9) Welding.
(10) Internal wiring practices including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

b. In this specification:

(1) Marking.
(2) Finish, protective.
(3) Printed wiring boards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)
b. Inspection covered by subsidiary documents. (See 4.4)
c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)
(2) Quality conformance inspection of preparation for delivery. (See 4.13)

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4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish five (5) first article units of the complete Power Supply PP-6224( )/U.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II. Inspections 1 through 5 shall be performed first prior to any additional testing. Reliability inspection (11) shall be performed last on all units at the same time.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Parts, materials, and processes; general	3.3
Printed wiring boards and protective coating	3.4, 3.4.1
Marking	3.5
Finish, protective	3.6
Tropicalization	3.13

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TABLE II - First Article inspection

Inspection	Requirement Paragraph	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units.				
2. Preconditioning, bounce Burn-in	3.11 3.12	4.6.1 4.6.2	Inspections to be performed on all units.				
3. Group A Inspection	See Table III		Inspections to be performed on all units.				
4. Group B Inspection	See Table IV		Inspections to be performed on all units.				
5. Frequency-voltage-load matrix	3.7.1	4.7.1.2	Inspection to be performed on all units.				
6. Temperature Test Temperature Coefficient Relative Humidity	3.8.1 3.7.12 3.8.2	4.8.1 4.7.10 4.8.2	X X X				
7. Elevation Stability Electromagnetic Compatibility	3.8.3 3.7.11 3.9	4.8.3 4.7.9 4.11		X X X			
8. Immersion Blowing Sand and Dust	3.8.4 3.8.7	4.8.4 4.8.9			X X		
9. Vibration Bounce Shock, Ballistic	3.8.5.1 3.8.5.2 3.8.5.2	4.8.5 4.8.6 4.8.7				X X X	
10. Fungus	3.8.6	4.8.8					X
11. Reliability	3.15	4.12	Inspection to be performed last on all units.				

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4, 4.6 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE III - Group A inspection

Inspection	Reqt. Para.	Insp. Para.	AQL dphu 1/	
			Major	Minor
Visual & mechanical	3.16	4.10	1.5 dphu	6.5 dphu
Input voltage	3.7.1	4.7.1)		
Output voltage	3.7.2	4.7.1)	Combined	
Power changeover	3.7.6	4.7.5)	1.5%	2/
Ripple voltage	3.7.3	4.7.2)		
Efficiency	3.7.9	4.7.8	1.5%	
Air-seal	3.14	4.8.10	1.5%	
Battery discharge	3.7.8.1	4.7.6	1.5%	
Protection, general	3.7.8.2	4.7.11	1.5%	
Remote sensing	3.7.10	4.7.7	1.5%	
Overload protection	3.7.5	4.7.4	1.5%	
Current limiting	3.7.7	4.7.4	1.55%	

1/ dphu - Defects per hundred units.
2 These inspections are major category only.

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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table IV shall be as follows:

AQL	INSPECTION LEVEL
4.0 dphu	S-4
6.5 dphu	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

Table IV - Group B inspection

Inspection	Reqt. Para.	Insp. Para.	AQL
Recovery time	3.7.4	4.7.3	4.0%
Interchangeability	3.10	4.9	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

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TABLE V - Group C inspection

Inspection	Reqt. Para.	Insp. Para.
Group 1		
Frequency - voltage - load matrix	3.7.1	4.7.1.2
Immersion	3.8.4	4.8.4
Group 2		
Stability	3.7.11	4.7.9
Temperature Test	3.8.1	4.8.1
Temperature Coefficient	3.7.12	4.7.10
Group 3		
Elevation	3.8.3	4.8.3
Relative Humidity	3.8.2	4.8.2
Vibration	3.8.5.1	4.8.5
Bounce	3.8.5.2	4.8.6
Group 4		
Shock; ballistic	3.8.5.2	4.8.7
Electromagnetic compatibility	3.9	4.11
Fungus	3.8.6	4.8.8
Blowing sand and dust	3.8.7	4.8.9

4.5.3.1 Group C sampling.

4.5.3.1.1 Sampling for inspection (Group 1). One sample shall be selected at random, from every 50 equipments or fraction thereof produced and subjected to each inspection specified. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

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4.5.3.1.2 Sampling for inspection (Group 2). Two samples shall be selected at random each month and subjected to each inspection specified. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.3 Sampling for inspection (Group 3). One sample shall be selected at random, from each 200 equipments produced or fraction thereof, and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

4.5.3.1.4 Sampling for inspection (Group 4). This inspection shall be required only when first article samples are not specified by contract. When required, one sample shall be selected at random, at the start of the contract, from the first quality conformance inspection production lot (See 4.5.3), and subjected to each inspection specified.

4.5.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection and the frequency-voltage-load matrix inspection of 4.7.1.2 (See Group I, Table V).

4.5.4 Group D inspection. This inspection shall consist of the test specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VI - Group D inspection

Inspection	Reqt. Para.	Insp. Para.
Reliability	3.15	4.12

4.5.4.1 Group D failures. In the case of a Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

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4.5.4.2 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and passed Group A inspection.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick-strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After the bounce preconditioning, the unit shall not be repaired, alined, or otherwise changed prior to subjection to acceptance inspection. (See 3.11)

4.6.2 Burn-in. Each equipment submitted for acceptance shall be subject to a burn-in period with 24 hours of continuous failure free on time. The test shall be conducted at room ambient temperature using an input of 115 Vac, 60 Hz and an output of 28 Vdc and 25 amperes. If a failure occurs, the equipment shall be repaired and returned to testing until it has accumulated 24 hours of continuous on time without a relevant failure. All failures shall be recorded and analyzed. A relevant failure for burn-in purposes shall be defined as the inability of the equipment to meet the requirements of this specification. (See 3.12)

4.7 Electrical tests.

4.7.1 Voltage tests, input and output.

4.7.1.1 Input and output. The output voltage shall meet the requirements of 3.7 over the load current range 0 to 25 amperes and over the input voltage range 103.5 to 126.5 volts/207 to 253 volts, 60 Hz.

4.7.1.2 Frequency-voltage-load matrix test. The output voltage shall meet the requirements of 3.7.2 when the load, input voltage, and frequency are varied as specified below; also when the external control is adjusted over the output voltage range:

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Input voltage (ac)	Freq (Hz)	Load (amps)
103.5	47.5, 63, 420	0 to 25
115	47.5, 63, 420	0 to 25
126.5	47.5, 63, 420	0 to 25
207	47.5, 63, 420	0 to 25
230	47.5, 63, 420	0 to 25
253	47.5, 63, 420	0 to 25

4.7.2 Ripple voltage. The rms ripple voltage in the dc output shall be measured with an electronic voltmeter (Ballantine Model 300 or equal). The peak to peak ripple voltage shall be measured with a calibrated oscilloscope having a bandwidth of 20 Hz to 10 MHz (Tektronix Model 564 with 3A1 amplifier unit, or equal). This test shall be made using the following input voltages: 103.5 Vac - 60 Hz, 115 Vac - 60 Hz and 126.5 Vac - 60 Hz. (See 3.7.3)

4.7.3 Recovery time. With the power supply operating from 115 Vac - 60 Hz, the load shall be switched from 25 to 0 amperes and the power supply switched on and off at each of the load conditions. The voltage transients shall be noted on a calibrated oscilloscope. This procedure shall be repeated when the load is switched from 0 to 25 amperes. The transient voltage shall meet the requirements of 3.7.4.

4.7.4 Overload protection and current limiting. With the power supply operating at full load (25 amps) the load shall be increased gradually to a short circuit condition and the output current noted during the current limiting condition. Upon removal of the short, the power supply shall automatically return to normal operating conditions. Repeat five (5) times. The power supply shall then meet the requirements of 3.7.1, 3.7.2 and 3.7.3. This test to be performed at room temperature and during the temperature tests of 4.8.1. The input voltage during the tests shall be 115 V rms, 60 Hz. Visual examination shall be made for any physical damage that might have occurred due to the overload conditions. (See 3.7.5 and 3.7.7)

4.7.5 Power changeover. Using a 115 V, 60 Hz single phase variable supply, reduce the AC input voltage until the power supply automatically changes over to DC and note the AC voltage at this point. Increase the AC input voltage until the power supply automatically changes back to AC and note the value of AC input voltages at

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this point. These two voltages shall meet the requirements of 3.7.6. Repeat this test using 115 V - 400 Hz input and 230 V - 60 Hz input and 230 V - 400 Hz input. This test shall be conducted with both a 2 amp and a 25 amp load for each input voltage and frequency specified. Proper "light" indication shall be checked for conformance to requirements of 3.7.6. Under the power changeover condition, indicated when the red light on the front panel labelled "DC EMER" is on, a check shall be performed to determine the continuity and consistency of polarity of the circuit connecting the "battery in" plug and the "power out" plug.

4.7.6 Battery discharge. With the input power source removed from the power supply, a fully charged 24 volt secondary type battery shall be connected across the dc output in correct polarity. The battery shall then be connected in reverse polarity. The reverse current flow from the battery for either polarity shall not exceed ± 100 milliampere. (See 3.7.8.1)

4.7.7 Remote sensing. With the power supply adjusted for remote sensing, the load shall be connected through a 25 foot cable, 4 conductor #12 wire size (2 conductors for sensing). The output voltage shall be measured at the load and shall meet the output voltage requirements of 3.7.2 (except the voltage range may be limited to 24 to 29 Vdc) when tested as specified in 4.7.1. During this test there shall be no evidence of unstable operation. (See 3.7.10)

4.7.8 Efficiency. The overall efficiency of the power supply shall be measured under conditions of full load, 28.0 volts at 25 amperes, and at an input of 115 volts rms, and at power frequencies of 50, 60 and 400 Hz. The percent efficiency shall be determined by the following and shall meet the requirements of 3.7.9:

$$\frac{\text{Output power (volts x current x 100)}}{\text{Input power (measured with wattmeter)}} = \%\ \text{efficiency}$$

4.7.9 Stability. The power supply shall be operated from 115 volt 60 Hz source with a 25 ampere load at nominal output voltage for 24 hours at room temperature. At the end of this test, the power supply shall meet the requirements of 3.7.11. Repeat this test using an input of 115 Vac, 400 Hz.

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4.7.10 Temperature coefficient. The power supply shall be operated from 115 V 60 Hz source with a 25 ampere load at nominal output for 4 hours in an ambient temperature of 70°F (21.1°C). The ambient temperature shall then be increased to 120°F (49°C) and the equipment operated for 4 hours. At the end of this test, the change in the output voltage shall not have exceeded 1.0%. Repeat this test at an ambient temperature starting at -40°F (-40°C) and increasing to +10°F (-12.2°C). (See 3.7.12)

4.7.11 General protection. (See 3.7.8.2). With the AC input adjusted to 115 Vac, 60 Hz, adjust the DC output to 28 Vdc with a 25 ampere load. Quickly raise the input voltage to 139 Vac and then reduce it back to 115 Vac. At the end of the test, the power supply shall conform to the requirements of 3.7.2, 3.7.3 and 3.7.7.

4.8 Environmental conditions test. (See 3.8)

4.8.1 Temperature test.

4.8.1.1 Low temperature. The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -80°F (-62.2°C) and the operating temperature (Step 4) shall be -40°F (-40°C). Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.1.1.

4.8.1.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure I of MIL-STD-810. The operating temperature (Step 4) shall be +150°F (66°C). Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.8.1.2.

4.8.2 Humidity. The PP-6224( )/U shall be subjected to the humidity test specified in Method 507, Procedure II of MIL-STD-810, with test measurements during Step 4 of each cycle and the last 5-hour period of the fifth 48-hour cycle in Step 6. Inability of the equipment to meet the requirements of 3.8.2 shall constitute failure of this test.

4.8.3 Altitude (elevation). The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.8.3.

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4.8.4 Leakage (immersion) test. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure 1, of MIL-STD-810.

4.8.5 Test for vibration. The equipment shall be tested in accordance with Method 514, Procedure VIII, of MIL-STD-810 using Curve V. A resonance search and resonance dwell shall be conducted as part of the sinusoidal cycling for each axis. The equipment shall be vibrated for 1/2 hour at the most severe resonant frequency for that axis. The total cycling time, including resonance search and dwell, shall be three hours for each axis.

4.8.6 Bounce (vibration) test. The equipment shall be tested in accordance with Method 514, Procedure IX, of MIL-STD-810. Package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds, and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.8.7 Shock test; ballistic (See 3.7.8.2). The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The equipment shall be secured in its normal operating position to the test plate. The test shall consist of a total of 9 blows: One each 1 foot blow, 3 foot blow, and 5 foot blow, on the back, side, and top of the test plate with the blows being directed to the equipment respectively. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.8.8 Fungus. The power supply, with the top cover removed, shall be subjected to the fungus test specified in Method 508, Procedure 1, of MIL-STD-810. There shall be abundant growth colonization (6.8.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the power supply is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the power supply to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.9 Dust (fine sand) test. The equipment shall be tested in accordance with Method 510, Procedure 1, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.7.

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4.8.10 Air-seal test. The equipment shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily. (See 3.14)

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable subassemblies and replaceable parts, when such dimensions affect mating of parts.

b. Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

c. Location of fasteners on separate parts or assemblies which must mate together such as cases, covers, and mountings.

4.10 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252.

4.11 Electromagnetic interference (EMI). The equipment shall be tested for compliance with the requirements of 3.9 of this specification. Test set ups and procedures employed shall conform to the measurement techniques of Notice 3 of MIL-STD-462.

4.12 Reliability testing.

4.12.1 First Article. The Group D reliability test shall be conducted on five (5) samples.

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4.12.2 Quality conformance inspection. The Group D reliability tests shall be performed on the first production lot, only if not performed during First Article testing. The contractor shall conduct the reliability test on 10 units from the lot based on the test conditions of 4.12.3. A lot for reliability purposes shall be defined as either 100 units or three month's production whichever is larger.

4.12.3 Reliability test conditions. The reliability test shall be based on an MTBF of 5000 hours. The test shall be run according to the conditions of test level C of MIL-STD-781 except that no vibration shall be performed and equipment cycling shall be limited to 4 hours of on time per cycle. Accept-reject criteria shall be determined using test plan VI of MIL-STD-781. Each unit shall have an input of 115 VAC at 60 Hz and an output of 28 VDC and 25 amperes. A failure for reliability purposes shall be defined as the inability of any equipment to meet the requirements of this specification. Each unit shall be continuously monitored to detect any failures. The total test time for each equipment shall be in accordance with 4.2.2.4 of MIL-STD-781. Paragraphs 5.5 thru 5.9 of MIL-STD-781 shall apply.

4.13 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.13.1 Rough handling test (packaging). When the rough handling test in accordance with MIL-P-116 is required by contract (See 6.2.e) the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Insp. Para.	Measurement	Input voltage
4.7.1	Output voltage (No loan and full load)	115 Vac RMS, 60 Hz

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G0001.

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6. NOTES

6.1 Intended use. Power Supply PP-6224( )/U is intended to provide dc power for operation of Receiver-Transmitter RT-246( )/VRC-12 or RT-524( )/VRC-12 from a 115/230 volt, 50, 60, or 400 Hz single phase ac power source.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article inspection:

(1) Five sample units of each item cited in Section 1 are generally required. Lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on another sample unit. (See 3.1)

(2) Preproduction pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted.
Inspection to be performed thereon. (See 4.13.1)

c. Marking and shipping of samples.

d. Place of final inspection.

e. When rough handling test is required. (See 4.13.1)

f. Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.13)

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-6224M/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C and D inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2 and 4.5.4.1)

6.6 Color. The color chip furnished by the procuring agency (3.4) will match color chip No. X-24087 and may be obtained upon request to Commanding Officer, U. S. Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.7 Location of air-seal test. It is desirable that the airseal test (4.8.10) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.8.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

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6.8.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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